5 $\lim_{h\to 0} \frac{f(x_0+h) - f(x_0-h)}{h}$

$= \lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h} - \frac{f(x_0-h) - f(x_0)}{h}$

$= \lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h} + \frac{f(x_0-h) - f(x_0)}{-h}$

$= 2f'(x_0)$

9 (1)

$f(x) = \frac{1}{x} \quad (x \neq 0)$

$f'(x) = -\frac{1}{x^2}$

접선 : $-\frac{1}{a^2}(x-a) + f(a) - y = 0$

법선 : $\frac{a^2}{1}(x-a) + f(a) - y = 0$

10. $f(4) = 3$

$f'(4) = \frac{3-0}{4-2} = \frac{3}{2}$ //



최대 : 존재 X , 최소 : 존재 X

3. 1

2 (2) $f(x) = \frac{1}{x^3}$ $(x \neq 0)$

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{\frac{1}{(x+h)^3} - \frac{1}{x^3}}{h}$$

$$= \lim_{h \to 0} \frac{x^3 - (x+h)^3}{x^3 \cdot (x+h)^3} \cdot \frac{1}{h}$$

$$= -\frac{3x^2}{x^6}$$

$$= -\frac{3}{x^4}$$

4 (1) $\lim_{x \to 1} \frac{f(x^2) - f(1)}{x-1}$

$$= \lim_{x \to 1} \frac{f(x^2) - f(1)}{(x-1)(x+1)} \cdot (x+1)$$

$$= \lim_{x \to 1} \frac{f(x^2) - f(1)}{x^2 - 1} \cdot (x+1)$$

$$= 2f'(1)$$